Accidents That Changed the World

Day 1 · Reading (1 of 2)

What to do: Read it aloud, one paragraph each. Stop at every blue word: can anybody explain it?

Vocabulary topic: Science and experiments
20 new words (in blue) · EEW 1, Unit 2 · read about scientific accidents and talk about experiments



easy parts harder parts (most new words are here)

Scientists do **experiments** in a **laboratory**. They wear white coats, they work with strange **chemicals** and they write everything in a notebook. You think: science is a clean, careful job with a plan. But some of the best discoveries in history did not come from a plan. They came from a mistake. Here are three true stories.

In 1928, a doctor, Alexander Fleming, came back from his holiday to his laboratory in London. He forgot to clean one of his dishes with bacteria. When he looked at the dish, he saw a strange green mold, and around the mold the bacteria were dead. Fleming **approached** the dish and looked at it **carefully**. This mold could **kill** bacteria! Years later, the mold became penicillin, the first antibiotic. It saved millions of lives.

Fleming took a long holiday, and that is part of the story. He did not clean his dishes before he left. Sometimes a holiday is good for science!

The second story is about a hot chocolate bar. In 1945, an engineer, Percy Spencer, worked next to a machine that made radio waves. Suddenly the chocolate bar in his pocket melted. Spencer was **nervous** at first, but he was also curious. He put some popcorn near the machine: it made a **loud noise** and jumped out of the dish. Soon the whole laboratory had a lovely **smell** of popcorn. Then he tried an egg, and the egg exploded in the face of a colleague. It was a **terrible** day for his colleague, but a great day for the kitchen: the microwave oven was born.

Today almost every home has a microwave. It heats food in a few minutes. But the first one was as tall as a man, and it cost a lot of money.

Accidents That Changed the World

Day 1 · Reading (2 of 2)

What to do: After reading: which paragraphs were easy, which were hard? Which blue words did you know already?

The third story is about glue. In 1968, a scientist, Spencer Silver, tried to **create** a very strong glue. He failed: his glue was weak, and it was **worse** than every other glue in the laboratory. Silver was not angry. He thought the glue was a **secret** treasure, but for years nobody wanted it. Then his colleague Art Fry used the weak glue on small pieces of paper to mark the songs in his church book. The paper stayed, and then it came off easily, without any damage. The Post-it note was born.

What do these stories teach us? Three easy things. First, look carefully at your mistakes. Second, ask questions. Third, do not **shout** "It's a mistake!" too fast. A **project** can go wrong in a good way.

Science is also an **adventure**, and sometimes it can **scare** people. When the first scientists talked about invisible bacteria, many people **laughed** at them. They said it was impossible. Some people believed that sickness came from **evil** spirits. Today nobody laughs, because the scientists looked, listened and did not stop.

Here is a fun fact: in most laboratories there is a rule: no food and no drinks. The reason is simple. A chemical can look like water and be dangerous. That is why scientists always wear goggles, a white coat and gloves, and they never, ever taste anything.

Not every discovery was lucky. In 1864 there was a terrible explosion in the factory of the Nobel family near Stockholm. Five people died, and one of them was Alfred Nobel's younger brother. Alfred looked for a safer way to use the dangerous liquid, and he created dynamite. It was safer to carry, but people used it for war, too. At the end of his life, Nobel left his money for a prize for the best scientists, writers and peacemakers: the Nobel Prize.

So scientists must be careful and brave at the same time. They must be careful, because experiments can be dangerous. They must be brave, because the next discovery can be very strange.

So the next time something goes wrong in your kitchen, your school or your work, wait a moment before you throw it away. Maybe it is the next big discovery. Look at it, smell it, think about it, and ask: what can I do with this?

The 20 words

Day 1 · Vocabulary table · Topic: Science and experiments

What to do: Book: 4000 EEW 1, Unit 2, pages 14–17. Copy the words you do not know into the copybook.

Word	Meaning	По-русски	Example
adventure n.	an exciting or dangerous experience	приключение	Camping in the mountains is a great adventure.
approach v.	come near to something	приближаться, подходить	The cat approaches the bird slowly.
carefully adv.	with great attention, not fast	осторожно, внимательно	Read the question carefully.
chemical n.	a substance used in or made by science	химическое вещество	Don't touch these chemicals!
create v.	make something new	создавать	Artists create beautiful things.
evil adj.	very bad, cruel	злой, злобный	The evil queen in the story is very cruel.
experiment n.	a scientific test to find something out	опыт, эксперимент	We do an experiment in science class.
kill v.	make a living thing die	убивать	This soap kills germs.
laboratory n.	a room where scientists work	лаборатория	Scientists work in a laboratory.
laugh v.	make the sound that shows you think something is funny	смеяться	We laugh at funny videos.
loud adj.	with a lot of sound	громкий	The music is too loud!
nervous adj.	a little afraid or worried	нервный, взволнованный	I am nervous before a test.
noise n.	a sound, often a loud or bad one	шум	What is that noise?
project n.	a piece of work with a plan	проект	Our class project is about the sea.
scare v.	make someone afraid	пугать	Big dogs scare my little sister.
secret n./adj.	something that only a few people know	секрет; секретный	Can you keep a secret?
shout v.	say something very loudly	кричать	Don't shout in the library!
smell v./n.	what you notice with your nose	пахнуть; запах	The flowers smell nice.
terrible adj.	very bad	ужасный	The weather is terrible today.
worse adj.	more bad than before	хуже	My cold is worse today than yesterday.

More on the topic

Day 1 · Topic: Science and experiments · more words

What to do: Choose 5 collocations to learn. Use the reserve words when a student already knows the main ones.

Collocations: words that go together

do an experiment	проводить эксперимент
work in a laboratory	работать в лаборатории
make a loud noise	громко шуметь
shout at someone	кричать на кого-то
a terrible mistake	ужасная ошибка
keep a secret · tell a secret	хранить тайну · рассказать секрет
go on an adventure	отправиться в приключение
create a new project	создать новый проект
smell of smoke	пахнуть дымом
laugh at someone	смеяться над кем-то
approach the door carefully	осторожно подойти к двери
feel nervous about…	нервничать из-за…

Phrasal verbs

find out	узнать, выяснить	The scientists found out the answer by accident.
come up with	придумать	She came up with a clever idea.
work out	разобраться; вычислить	I cannot work out this problem.
turn on / turn off	включить / выключить	Turn off the machine, please.
try out	испытать, попробовать	Let's try out the new method.
blow up	взорвать(ся)	The egg blew up in the microwave.

Reserve words: for students who know the 20 already, or if you prefer another word

test	тест; проверять	**machine**	машина, аппарат
discover	открывать	**curious**	любопытный
invent	изобретать	**danger**	опасность
mistake	ошибка	**explode**	взрываться
result	результат	**goggles**	защитные очки

What to read next · 4000 Essential English Words

This week: **Book 1, Unit 2, "The Laboratory"**, pages 14–19. Next: **Unit 3, "The Report"**, pages 20–25. About experiments and fear: **Unit 1, "The Lion and the Rabbit"**, pages 8–13.